

The use of AI Generation report

Overview of the tool(s) used

In my latest project primarily involving concurrency programming, I employed the services of Generative AI (GenAI) tools, specifically ChatGPT, powered by OpenAI's GPT-4 architecture. This AI-driven tool is designed to understand and generate human-like text based on the input it receives. The capabilities of ChatGPT extend beyond simple text generation. From analysing code, offering algorithmic solutions, assisting in debugging, to providing detailed explanations of complex concepts.

Integration ChatGPT into workflow

The integration of GenAI into my workflow was seamless and multifaceted. At the initial stages of my project, I used ChatGPT to plan my next moves regarding the assignment. Listing out the primary objectives with steps needed to be taken has significantly boosted my productivity and fastened the progress of my project. By acknowledging which stage I was currently at, it also provides a sense of knowing what needs to be done next, reinforcing my path to completion of the project. While working on the project, chatGPT was also used to clarify my misunderstood concept since I was struggling to differentiate actual concurrency and printing out to the console one by one. In the end, it was used for debugging purposes, writing comments for every step to demonstrate the work of every step in my `run()` method.

Quality of the response and its impact on the solution

Although the genAI sometimes can overlook potential problems arising from changes when providing suggestions, the overall quality and correctness of generated texts from ChatGPT was personally consistent. ChatGPT was especially helpful when it comes to optimising and shortening my code. It analyses long paragraphs of codes, providing an example of how it could be shortened if it could be written more elegantly, resulting in neat, refined code, removing unnecessary code. From time to time, it also correctly detects faulty logic that I overlooked, answering the question why my code was not running as expected.

Improvement in Learning Experience and Problem-Solving Skills

Using GenAI tools has created a decent impact on problem-solving skills. The provided solution was often not determined enough, but elaboration further in the topics or

examples can always be provided. Posing questions more frequently also intrigued my curiosity on the topic, dwelling me into this interactive learning process, allowing me to bridge gaps in my knowledge more effectively. Additionally, the GenAI tool also provided different approaches to a certain problem, therefore giving me more point of view on the encountered challenge, meaning I can freely choose which approach could be more effective, more suitable for the project. I was able to develop a clearer understanding of the overall structure and logic required to implement a solution.

Challenges Faced and Solutions

One of the main challenges I faced while using GenAI tools was the occasional difficulty in interpreting code-related suggestions. Since the AI generates responses based on patterns learned from vast datasets, there were times when the suggestions were not entirely aligned with the specific requirements of my project. Another challenge was ensuring that I remained an active participant in the problem-solving process rather than relying too heavily on the AI. To address this, I made it a point to understand and analyse each suggestion before implementing it, using the tool as a guide rather than a crutch. Lastly, the genAI can misinterpret my code and my intentions from time to time. I learned to frame my queries more precisely and provide context to ensure that the AI understood my requirements fully.

Productivity and Time Saved

As mentioned above, the chatGPT tool helped me with task organisation and elaboration, which saves me time recalling my progression every time I get back to doing this assignment. In case of progression of my project, chatGPT also helps with pointing out where met exceptions could arise from, suggesting what can be improved to ensure an error free output. This puts out a large portion of time needed for debugging codes, gets me concentrating on the correction of code, optimising the run.

Ethical Considerations

While the use of GenAI tools provided numerous benefits, it also raised certain ethical considerations. One major concern is the potential for over-reliance on AI-generated solutions, which could lead to a diminished understanding of fundamental concepts. To mitigate this, I consciously make sure that seeking suggestions from ChatGPT shall only be the last approach in order to resolve a problem. By fully grasping the reasoning behind each suggestion, I believe this can be seen as a learning opportunity. Additionally, there is the question of data privacy. While using GenAI tools, I made sure not to input any sensitive or personal information, understanding that the AI processes data based on large, anonymized datasets.

Conclusion

In conclusion, the utilisation and integration of chatGPT into my project has been a highly positive experience. The tools have a good impact on my learning process while also helping with the practice stage. It also boosts my productivity with plans and comments. However, it is important to use the tool ethically, ensuring that they complement rather than replace traditional learning methods. By striking this balance, I was able to make the most of what GenAI has to offer, leading to a successful and enriching project outcome.